

NETWORK+ MASTER ACRONYM TABLE

Acronyms with checkmarks are not already covered in A+, new to Net+

Networking Fundamentals

Acronym	Expanded	Definition
IP	Internet Protocol	Assigns addresses to devices and ensures data is sent to the correct destination across a network.
TCP	Transmission Control Protocol	Reliable protocol that ensures data is delivered completely and in order.
UDP	User Datagram Protocol	Fast protocol that sends data without guaranteeing delivery.
MAC	Media Access Control	Unique hardware address assigned to a network interface.
✓ MTU	Maximum Transmission Unit	Largest packet size that can be sent over a network.
✓ TTL	Time to Live	Limits how long a packet can travel before being discarded.
✓ ARP	Address Resolution Protocol	Maps IP addresses to MAC addresses on a local network.
✓ ICMP	Internet Control Message Protocol	Used for diagnostics and error reporting (like ping).
✓ IGMP	Internet Group Management Protocol	Manages multicast group memberships.
✓ OSI	Open Systems Interconnection	7-layer model describing how data moves across a network.

Network Types & Infrastructure

Acronym	Expanded	Definition
LAN	Local Area Network	Network in a small area like a home or office.
WAN	Wide Area Network	Connects multiple LANs across large distances.
WLAN	Wireless Local Area Network	LAN that uses Wi-Fi instead of cables.
VLAN	Virtual Local Area Network	Logically separates networks on the same hardware.
✓ VPC	Virtual Private Cloud	Private network hosted within a cloud provider.
✓ VRF	Virtual Routing and Forwarding	Allows multiple routing tables on one device.
✓ VXLAN	Virtual Extensible LAN	Extends VLANs across large networks.
✓ SDN	Software-defined Network	Network controlled by software instead of hardware.
✓ SD-WAN	Software-defined Wide Area Network	Uses software to optimize WAN traffic routing.
✓ MPLS	Multiprotocol Label Switching	Speeds up traffic using labels instead of IP lookups.

Routing & Switching

Acronym	Expanded	Definition
✓ BGP	Border Gateway Protocol	Routes traffic between large networks on the Internet.
✓ OSPF	Open Shortest Path First	Finds the shortest path for data in a network.
✓ RIP	Routing Information Protocol	Uses hop count to determine best route.
✓ EIGRP	Enhanced Interior Gateway Routing Protocol	Cisco routing protocol for efficient routing decisions.
✓ IS-IS	Intermediate System to Intermediate System	Routing protocol used in large service provider networks.
✓ STP	Spanning Tree Protocol	Prevents loops in switched networks.
✓ RSTP	Rapid Spanning Tree Protocol	Faster version of STP.
✓ LACP	Link Aggregation Control Protocol	Combines multiple links into one connection.
✓ FHRP	First Hop Redundancy Protocol	Provides backup gateways.
✓ HSRP	Hot Standby Router Protocol	Cisco redundancy protocol for routers.
✓ VRRP	Virtual Router Redundancy Protocol	Standard protocol for gateway redundancy.

IP Addressing & DNS

Acronym	Expanded	Definition
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses.
DNS	Domain Name System	Translates domain names into IP addresses.
✓ DNSSEC	Domain Name System Security Extensions	Secures DNS from spoofing attacks.
✓ DoH	DNS over HTTPS	Encrypts DNS queries using HTTPS.
✓ DoT	DNS over TLS	Encrypts DNS queries using TLS.
✓ CIDR	Classless Inter-domain Routing	Flexible IP addressing method.
✓ VLSM	Variable Length Subnet Mask	Allows different subnet sizes.
NAT	Network Address Translation	Converts private IPs to public IPs.
✓ PAT	Port Address Translation	Allows many devices to share one public IP.
✓ IPAM	Internet Protocol Address Management	Tracks and manages IP addresses.
✓ FQDN	Fully Qualified Domain Name	Complete domain name (e.g., www.google.com).
✓ CNAME	Canonical Name	Alias for another domain name.
✓ MX	Mail Exchange	Identifies email servers.
✓ PTR	Pointer Record	Maps IP to domain name.
✓ NS	Name Server	Identifies DNS servers for a domain.
✓ SOA	Start of Authority	Contains DNS zone authority information.

Security Concepts

Acronym	Expanded	Definition
✓ CIA	Confidentiality, Integrity, Availability	Core goals of cybersecurity.
MFA	Multifactor Authentication	Uses multiple forms of verification.
NAC	Network Access Control	Controls who can access the network.
IAM	Identity and Access Management	Manages users and permissions.
DLP	Data Loss Prevention	Prevents sensitive data leaks.
✓ ZTA	Zero Trust Architecture	Trust no device by default.
UTM	Unified Threat Management	Combines multiple security tools.
✓ SASE	Secure Access Service Edge	Cloud-based networking and security.
✓ SSE	Security Service Edge	Cloud-delivered security services.
✓ WAF	Web Application Firewall	Protects web apps from attacks.

Security Protocols

Acronym	Expanded	Definition
AES	Advanced Encryption Standard	Strong encryption standard used globally.
✓ IPsec	Internet Protocol Security	Encrypts network traffic.
✓ AH	Authentication Header	Provides authentication for IP packets.
✓ ESP	Encapsulating Security Payload	Encrypts and authenticates IP traffic.
✓ SSL	Secure Sockets Layer	Older encryption protocol.
✓ TLS	Transport Layer Security	Modern secure encryption protocol.
✓ PKI	Public Key Infrastructure	Manages encryption keys and certificates.
✓ PSK	Pre-shared Key	Shared password for encryption.
TKIP	Temporal Key Integrity Protocol	Older wireless encryption method.
WPA	Wi-Fi Protected Access	Secure wireless encryption standard.
WEP	Wired Equivalent Privacy	Weak, outdated wireless encryption.
✓ WPS	Wi-Fi Protected Setup	Simplifies Wi-Fi setup (less secure).

Security Monitoring

Acronym	Expanded	Definition
DoS	Denial of Service	Overloads a system to make it unavailable.
DDoS	Distributed Denial of Service	Attack from multiple systems.

✓ IDS	Intrusion Detection System	Detects suspicious activity.
✓ IPS	Intrusion Prevention System	Detects and blocks threats.
✓ SIEM	Security Information and Event Management	Collects and analyzes security logs.

Wireless Networking

Acronym	Expanded	Definition
SSID	Service Set Identifier	Name of a Wi-Fi network.
✓ BSSID	Basic Service Set Identifier	MAC address of access point.
✓ ESSID	Extended Service Set Identifier	Name for extended wireless network.
✓ RSSI	Received Signal Strength Indicator	Measures Wi-Fi signal strength.
✓ MIMO	Multiple Input Multiple Output	Uses multiple antennas for performance.
LTE	Long-Term Evolution	4G wireless communication standard.

Hardware & Cabling

Acronym	Expanded	Definition
✓ NIC	Network Interface Card	Connects device to network.
AP	Access Point	Provides wireless access.
WAP	Wireless Access Point	Same as AP.
✓ MDF	Main Distribution Frame	Primary network hub location.
✓ IDF	Intermediate Distribution Frame	Secondary network hub.
✓ DAC	Direct Attach Copper	Short-distance high-speed cable.
BNC	Bayonet Neill–Concelman	Coaxial cable connector.
LC	Local Connector	Fiber connector type.
SC	Subscriber Connector	Fiber connector type.
ST	Straight Tip	Fiber connector type.
✓ MPO	Multifiber Push On	Multi-fiber connector.
✓ UTP	Unshielded Twisted Pair	Standard Ethernet cable.
✓ STP	Shielded Twisted Pair	Cable with shielding.
✓ PoE	Power over Ethernet	Sends power over network cable.
✓ QSFP	Quad Small Form-factor Pluggable	High-speed transceiver.
✓ SFP	Small Form-factor Pluggable	Network interface module.

Systems & Interfaces

Acronym	Expanded	Definition
OS	Operating System	Software that manages hardware and apps.
GUI	Graphical User Interface	Visual interface (windows/icons).
✓ CLI	Command Line Interface	Text-based interface.
✓ API	Application Programming Interface	Allows software to communicate.
✓ IIS	Internet Information Services	Microsoft web server.
VNC	Virtual Network Computing	Remote desktop access tool.
✓ RDP	Remote Desktop Protocol	Microsoft remote access protocol.

Cloud & Virtualization

Acronym	Expanded	Definition
IaaS	Infrastructure as a Service	Cloud provides hardware resources.
PaaS	Platform as a Service	Cloud provides platform for apps.
SaaS	Software as a Service	Cloud provides software access.
✓ NaaS	Network as a Service	Cloud-based networking services.
VM	Virtual Machine	Virtual computer running inside a host.
✓ NFV	Network Functions Virtualization	Virtualized network services.
✓ IaC	Infrastructure as Code	Managing infrastructure via code.

Storage

Acronym	Expanded	Definition
✓ DAS	Direct Attached Storage	Storage directly connected to device.
NAS	Network Attached Storage	Storage accessible over network.
SAN	Storage Area Network	High-speed storage network.
✓ FC	Fibre Channel	High-speed storage protocol.

Hardware Components

Acronym	Expanded	Definition
CPU	Central Processing Unit	The brain of the computer.

PSU	Power Supply Unit	Supplies power to components.
UPS	Uninterruptible Power Supply	Backup power device.
✓ APU	Audio Processing Unit	Handles audio tasks.
✓ PDU	Power Distribution Unit	Distributes power in racks.

Reliability Metrics

Acronym	Expanded	Definition
✓ MTBF	Mean Time Between Failures	Average time before failure.
✓ MTTR	Mean Time To Repair	Average repair time.
✓ RTO	Recovery Time Objective	Max acceptable downtime.
✓ RPO	Recovery Point Objective	Max acceptable data loss.

Policies & Compliance

Acronym	Expanded	Definition
AUP	Acceptable Use Policy	Rules for system usage.
EULA	End User License Agreement	Software usage contract.
NDA	Non-Disclosure Agreement	Protects confidential data.
✓ MOU	Memorandum of Understanding	Agreement between parties.
SLA	Service-Level Agreement	Defines service expectations.
✓ GDPR	General Data Protection Regulation	EU privacy law.
✓ PCI DSS	Payment Card Industry Data Security Standard	Secures payment data.

Industrial & IoT

Acronym	Expanded	Definition
✓ IoT	Internet of Things	Smart connected devices.
✓ IIoT	Industrial Internet of Things	IoT used in industry.
✓ ICS	Industrial Control Systems	Controls industrial processes.
SCADA	Supervisory Control and Data Acquisition	Monitors industrial systems.
✓ OT	Operational Technology	Systems used in industrial environme